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Hydration Dynamics of Hydrocolloid Particles - Part 2: Molecular Dynamics Simulation and Correlation of Self-Diffusion Coefficients with Water Vapor Sorption Kinetics

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Abstract

In Part 1, experimental data showed that the vapor sorption rates of water for corn and wheat starches were reduced at high water activities (>0.4). Since water mobility in starch is not expected to decrease with increasing water activities, in the current study, we investigate the causes of this phenomenon through molecular dynamics (MD) simulations to predict the water sorption isotherms and self-diffusivity. It was shown that the self-diffusivity of water and the characteristic sorption parameter monotonically increase with increasing water activity (A_w). Three distinctive stages of water sorption were observed. At low values of A_w (<0.4), water molecules were mostly bound to the surface of the starch polymer, forming hydrogen bonds with the hydrophilic groups on the surface, and the water mobility was rather restricted by the rigid structures of starch. At intermediate values of A_w (0.4 to 0.6), water penetration into the bulk of the starch polymer was initiated resulting in swelling of the polymer chains and plasticization. Moreover, there was a corresponding increase in the characteristic sorption parameter and self-diffusivity. At high values of A_w (>0.6), the water self-diffusivity, and hence mobility, approached that of free water, and the characteristic sorption parameter continued to increase due to swelling. An increase of bifurcated (double) water-polymer hydrogen bonds was observed with increasing A_w that stabilised the starch structure. A more rigorous Darcy theory of unsaturated moisture transfer in porous media was used to obtain water permeability values from the experimental data that were reported in Part 1. The calculated water permeabilities increased monotonically with increasing A_w and correlated closely with the self-diffusivity of water following a simple exponential relationship. The predictions of the MD simulations agreed closely with the data reported in Part 1.

Keywords: Starch Hydration, Water Permeability, Non-Fickian Diffusion Model, Starch Free Volume

1. Introduction

Starch is a type of hydrocolloid found in most plants that is composed primarily of two types of glucose polymers, amylose, and amylopectin, joined by glycosidic bonds. Amylose is a linear molecule with α -(1,4)-linked glucose units. Amylopectin is made up of α -(1,4)-linked chains connected by α -(1,6) linkages (Bertoft, 2017). The ratio of amylose to amylopectin differs for various starch sources and dictates its properties and functionality (Zhong et al., 2023). Starch is widely used in industry, particularly for foods and pharmaceuticals. Understanding the hydration behaviour of starch is essential for controlling the properties in the various applications such as the thickening of aqueous formulations.

The hydration of starch is a complex process that involves multiple mechanisms, including surface adsorption, capillary condensation, and bulk absorption (Le Thanh-Blicharz et al., 2021). In Part 1, it has been shown that the measured kinetics of water vapor sorption of corn and wheat starch can be accurately described by the sigmoidal rather than the Fickian diffusion model. This suggests that the water vapor sorption is governed by first-order kinetics of the surface sorption coupled with bulk Fickian diffusion. It was also shown that initially the moisture diffusivity increased with increasing A_w and reached a maximum value around $A_w = 0.3$ after which it decreased. The reduction in the bulk diffusivity with increasing water activities at high hydration levels is a rather unusual behaviour since the water mobility in starch should be expected to increase with increasing values of the water activity (A_w). Hence, in this study (Part 2), the molecular-level interpretation of this behaviour is investigated using molecular dynamics (MD) simulations of corn starch molecules consisting of multiple amylose and amylopectin units.

MD simulations were conducted at different hydration levels to predict the values of A_w and the self-diffusion coefficient. They were initially validated by comparing the predicted water sorption isotherm with the experimental data of Part 1. Using the Darcy theory of unsaturated moisture transfer in porous media, a more rigorous constitutive analysis (Weerts et al., 2005) also was conducted to show that the bulk diffusion coefficient of moisture can be related to water permeability and retention capability of starch. The derived water permeability increased monotonically with increasing A_w and correlated closely with the self-diffusion coefficient of water predicted by the MD simulations, suggesting that water mobility in the hydrocolloid system increases with hydration level. The analysis of the MD simulation results suggests that the observed decrease in the bulk diffusion coefficient at higher water activities is due to the increase in water sorption capacity, leading to a longer time required for water vapor sorption to reach equilibrium. The molecular basis, as predicted by the radius of gyration of the polymer, hydrogen bonds, bound water, and free water, also is consistent with the increased characteristic sorption parameter and mobility with increasing hydration levels.

2. Materials and Methods

2.1 Molecular dynamics (MD) simulation

MD simulations were performed to obtain the sorption isotherm and self-diffusion coefficient of water at different water activities (A_w) for corn starch molecules. The initial molecular structures of corn starch were generated using the CHARMM-GUI (Guterres et al., 2022). This was followed by Grand Canonical Monte Carlo (GCMC) insertion and deletion moves of water molecules in the corn starch systems. These moves were performed to sample the number of water molecules in the system and achieve equilibrium at the desired chemical potential which is related to the A_w . OpenMM (Eastman et al., 2017) was used to perform

GCMC moves and obtain the starch-water systems at different A_w values. All the MD simulations were performed using GROMACS 2021.4 (Pronk et al., 2013) together with the CHARMM36 force field (Huang et al., 2017) and the TIP3P water model (Mark & Nilsson, 2001), which is consistent with the CHARMM36 force field parameterization.

2.1.1 Starch representation and hydration levels

The corn starch molecules were modelled as a combination of amylose (5%) and amylopectin (95%) chains in their native state. The doGlycans tool (Danne et al., 2017) was used to prepare the initial carbohydrate structures. Simulated amylose molecules were composed of 10 α -D-glucose units connected via $\alpha(1\rightarrow4)$ glycosidic bonds. Amylopectin consisted of 15 α -D-glucose units linked linearly via $\alpha(1\rightarrow4)$ glycosidic bonds. The branching in amylopectin occurred at two locations (the 5th and 10th α -D-glucose units) through $\alpha(1\rightarrow6)$ glycosidic bonds. The initial structures reflect the natural arrangement of amylose and amylopectin in unprocessed corn starch. The molecular structures of the simulated glucose, amylose, and amylopectin are shown in Fig. 1.

The starch molecules were solvated in a simulation box using the TIP3P water model. Hydration levels were simulated by controlling the number of water molecules in the system to achieve specific A_w levels. It allows water molecules to be inserted and deleted from the system according to the desired chemical potential. Fig. 2 shows the procedure followed for the addition of water molecules to the starch molecules in the simulation box.

2.1.2 Simulation protocol

The system was subjected to energy minimization using the steepest descent algorithm to achieve quasi-equilibrium at a lower-energy configuration. This was followed by two stages of equilibration: Constant Number, Volume, and Temperature (NVT) and Constant Number, Pressure, and Temperature (NPT) ensemble equilibration in succession. The NVT ensemble was carried out at 313.15 K with the V-rescale thermostat. Location restraints were applied to the starch molecules to allow the water molecules to equilibrate in close proximity, with the restraints being gradually reduced until equilibration was achieved (200 ns). The NPT ensemble was implemented at 313.15 K and 1 bar, using the Nosé-Hoover thermostat and the Parrinello-Rahman barostat, for 100 ns.

Equilibrium was determined by monitoring the stabilization of key thermodynamic and structural parameters over time. The total energy, temperature, and density of the system were tracked throughout the simulation, and equilibrium was confirmed when these properties fluctuated around a constant mean value. Moreover, the number of water molecules in the solvation shell remained consistent, suggesting that the system had reached a stable hydrated state.

For the production runs, simulations were conducted in the NPT ensemble for 100 ns at various A_w levels. The LINCS algorithm was used to constrain the bond lengths, with a 2 fs time step. The Particle Mesh Ewald (PME) was employed for long-range electrostatic interactions with a real-space cutoff of 1.2 nm. van der Waals interactions were truncated at 1.2 nm with a switch function starting at 1.0 nm.

2.1.3 Hydrogen bond vibration analysis

The MD simulations were validated by analysing the behaviour of the hydrogen bond displacements. Due to the noise, pre-processing steps were required to accurately identify the oscillation period. The detailed procedure is described below.

The raw displacement data of the hydrogen bonds from the simulation were found to contain noise due to molecular fluctuations as shown in Fig. 3a. To reduce random fluctuations, the displacements were first centred around zero as shown in Fig. 3b. A moving average filter with a specified window size of 20 was then applied as shown in Fig. 3c. This smoothing technique reduced high-frequency noise. The filtered signal allowed for more reliable zero-crossing detection.

The oscillation period was determined by identifying the time intervals between successive zero-crossings in the smoothed displacement data. Zero-crossings were defined as points where the displacement crossed the zero line, which transitioned from positive to negative or vice versa. The time intervals between every second zero-crossings were then calculated to determine the duration of each oscillation cycle as shown in Fig. 3d. These intervals were used to calculate the individual oscillation periods. Due to the noisy nature of MD simulations, slight variations in the oscillation period were observed. To obtain a more reliable estimate, the average period t_{osc} was calculated by taking the mean of these intervals and was calculated to be 9.76 fs.

The average oscillation period was converted into its corresponding wavenumber using the relation:

$$\tilde{\nu} = \frac{1}{t_{osc}} \times \frac{1}{c} \quad (1)$$

where c is the speed of light. The calculated wavenumber of 3417 cm^{-1} was found to be in the expected range of literature values for hydrogen bond vibrations ($3,200\text{--}3,650 \text{ cm}^{-1}$) (Ikemoto et al., 2022).

2.1.4 Calculation of bound water

A two-step approach that incorporates both structural and temporal criteria was used to identify bound water molecules. For the structural component, hydrogen bonds between water molecules and the starch polymer chains were detected based on a distance criterion: a hydrogen bond was considered to be present when the distance between the donor (starch oxygen atom) and acceptor (water oxygen atom) was $\leq 3.5 \text{ \AA}$. For the temporal component, water molecules were classified as bound only if they maintained these hydrogen bonds for more than 50% of the total simulation time. This temporal filter distinguished persistently bound water molecules from those making transient interactions with the starch polymer. Water was considered to be bound when both structural and temporal criteria were satisfied.

2.1.5 Water activity calculation using the test particle insertion (TPI) method

To calculate the water activity, A_w , the test particle insertion (TPI) method was employed, which estimates the chemical potential of the water molecules in the system. Water activity is defined as:

$$A_w = \frac{p_{System}}{p_{Pure}} \quad (2)$$

where p_{System} and p_{Pure} are the vapor pressures of water in the system and pure water, respectively.

In an MD simulation, the water activity is related to the chemical potential ψ :

$$A_w = \exp \{(\psi_{System} - \psi_{Pure})/RT\} \quad (3)$$

where ψ_{System} is the chemical potential of water in the simulated system, ψ_{Pure} is the chemical potential of pure water, R is the gas constant and T is the absolute temperature. The TPI method involves inserting a virtual water molecule (test particle) at random positions in the system and calculating the total interaction energy between this inserted particle and the remainder of the molecular system. The excess chemical potential $\Delta\psi_{excess}$ is given by:

$$\Delta\psi_{excess} = -k_B T \ln \langle \exp(-U_{test}/k_B T) \rangle \quad (4)$$

where U_{test} is the potential interaction energy between the test particle and the system, k_B is the Boltzmann constant, T is the temperature, $\langle \dots \rangle$ represents averaging over the test particle insertions. This equation is based on the Widom insertion method (Widom, 1963), which uses statistical averaging to estimate the energy required to add a water molecule to the system. By averaging over multiple test particle insertions at different points in the system, the excess chemical potential $\Delta\psi_{excess}$ of water is obtained. This is the additional energy required to add a water molecule to the system compared to adding it to pure water.

The values of $\psi_{System} - \psi_{Pure}$ were obtained from a separate simulation that contained pure water only. The relationship between A_w and the RH is given by:

$$RH = A_w \times 100 \quad (5)$$

2.1.6 Calculation of the water adsorption isotherm

To calculate the water adsorption isotherm from the MD simulations, the water activity was controlled by varying the number of water molecules in the system using the Grand Canonical Monte Carlo (GCMC) approach, as discussed in *Section 2.1.1*. The amount of water adsorbed was determined by calculating the mass of water retained within the starch-water system at equilibrium for each value of the A_w . The simulation was allowed to equilibrate in order to ensure that the water molecules reached an equilibrium state in terms of the interactions with each other and the starch matrix. At equilibrium, the water content was measured by counting the number of water molecules within the simulation box. The mass of water molecules absorbed is expressed as the wet mass percentage, $W_{absorbed}$ (%):

$$W_{absorbed} (\%) = \frac{m_{water}}{m_{system}} \times 100 \quad (6)$$

where m_{water} is the total mass of water in the system after equilibration and m_{system} is the total mass of the starch molecules and water in the system after equilibration.

2.1.7 Self-diffusion coefficient from MD simulation

The self-diffusion coefficient, D_{self} , of the water molecules was calculated using the Einstein relationship, which relates the mean square displacement (MSD) of the molecules to time, t :

$$D_{self} = \lim_{t \rightarrow \infty} \frac{1}{6t} \langle [r(t) - r(0)]^2 \rangle \quad (7)$$

where $[r(t) - r(0)]^2$ is the MSD of water molecules

2.2 Unsaturated moisture transfer

As described in Part 1, Dynamic Vapour Sorption (DVS) was used to measure the water vapour sorption kinetics and isotherm of corn starch samples. The bulk diffusivity of moisture transfer was obtained by fitting the water sorption kinetics data with the sigmoid model for which surface sorption is the rate-limiting step prior to bulk diffusion. The water vapour sorption at the sample surface follows first-order kinetics:

$$C_s(t) = C_{\infty} [1 - \exp(-t/\tau)] \quad (8)$$

where $C_s(t)$ is the moisture content at the sample surface at time t , C_{∞} is the equilibrium moisture content and τ is a characteristic time. Moisture transfer from the surface into the bulk is described by the Fickian diffusion equation:

$$\frac{\partial C}{\partial t} = D_{bulk} \nabla^2 C \quad (9)$$

where ∇^2 is the Laplace operator, D_{bulk} is the bulk diffusion coefficient and C is the moisture content. Eqs (8) and (9) can be solved analytically for simple geometries such as spheres, cylinders and/or thin layer slabs, to derive the sigmoid model (Crank, 2011).

The Fickian equation for moisture transfer is limited to homogeneous media. A more rigorous approach to formulating unsaturated moisture transfer in porous media is to adopt the extended Darcy porous flow theory. This framework uses moisture potential as the driving force while accounting for the effects of capillary force, osmotic pressure, water absorption by the matrix, and hydraulic pressure. The rate of unsaturated moisture transfer in 1D is given by (Whitaker, 1986):

$$j_{water} = -K \frac{\partial \psi}{\partial x} \quad (10)$$

where j_{water} is the moisture flux, K is the moisture conductivity or permeability and $\partial \psi / \partial x$ is the gradient of the moisture potential.

For moisture transfer under DVS conditions, hydraulic pressure and gravity effects can be ignored. Setting the reference moisture potential of Eq. 3 at saturation as zero, Kelvin's equation (Campbell et al., 1993) can be recovered and used to relate the moisture potential to the A_w as follows:

$$\psi = \frac{RT \ln A_w}{gM} \quad (11)$$

where M is the molecular weight of water and ψ is the moisture potential.

The application of the extended Darcy theory of unsaturated moisture transfer (Van Genuchten, 1980) leads to the following mixed form of a mass balance equation of moisture migration in starch:

$$\frac{\partial C}{\partial t} = K \nabla^2 \psi \quad (12)$$

Substituting Eq. (11) into (12), the mass balance equation of unsaturated moisture migration from the surface to the bulk can be written as:

$$\frac{\partial C}{\partial t} = K \frac{RT}{gMA_w} \nabla^2 A_w \quad (13)$$

Noting the following chain rule from the moisture sorption isotherm relationship between the moisture content C and the water activity A_w :

$$\nabla^2 A_w = \frac{1}{\frac{dC}{dA_w}} \nabla^2 C \quad (14)$$

Eq. (13) can then be re-written as:

$$\frac{\partial C}{\partial t} = K \frac{RT}{gMA_w} \frac{1}{\frac{dC}{dA_w}} \nabla^2 C \quad (15)$$

Comparing Eqs (15) with (9) of Fick's law of diffusion for bulk moisture transfer, the following relationship is obtained:

$$D_{bulk} = K \left(\frac{RT}{gMA_w} \right) \left(\frac{dC}{dA_w} \right)^{-1} \quad (16)$$

Thus:

$$K = D_{bulk} \left(\frac{gMA_w}{RT} \right) \left(\frac{dC}{dA_w} \right) = D_{bulk} H \quad (17)$$

where $H = (gMA_w/RT)(dC/dA_w)$ is denoted as the characteristic sorption parameter and dC/dA_w is the gradient of the sorption isotherm. H represents the capacity of a material to retain water and is influenced by the structure of the material and the thermodynamic interactions between water molecules and the material matrix. H quantifies the ratio between A_w increase and the moisture content increase of a material.

Eq. (17) is used to derive the moisture permeability from the reported values of moisture diffusivity in Part 1.

3. Results and Discussion

3.1 Bound and unbound water dynamics

Hydrogen bonding between water molecules and starch can be formed as single bonds and bifurcated (double) bonds. Those that are bifurcated are formed when both –OH groups of

water act as hydrogen bond donors and acceptors. The formation of such bonds in water-starch systems creates a synergistic effect that results in a more stable network structure (Gao et al., 2021). Figs 4a and 4b show the simulation results of the number of single and bifurcated hydrogen bonded water molecules at different hydration levels respectively. The corresponding amount of bound and unbound water molecules are plotted in Fig. 4d.

Although the total number of hydrogen bonds formed between water molecules and starch polymers is continuously increasing with the water activity, three distinctive stages of water sorption can be observed. At values of $A_w < 0.4$, most absorbed water molecules form hydrogen bonds with the starch polymer chains. The number is approximately constant and there is a slight increase in the number of water molecules forming bifurcated hydrogen bonds with starch. There is a small increase in the total unbound water molecules which mostly contributes to the overall increase of absorbed water, indicating the exposed hydrophilic sites of starch surface appear to have been already occupied by water molecules at a relatively low A_w values of approximately 0.1. A transition occurred at approximately an A_w value of 0.4 where the number of bound water molecules further increased. This can be attributed to the increased water molecules penetrating the starch polymer, disrupting polymer-polymer hydrogen bonds and forming new water-polymer hydrogen bonds (Zhiguang et al., 2022). This is supported by the decrease of polymer-polymer hydrogen bonds shown in Fig 4c. At $A_w > 0.4$, a significant amount of free water molecules are absorbed and the increase in unbound water accelerated with increasing A_w . Like the first stage, the number of water molecules bound to the starch polymer stayed mostly constant and there was a small increase in the number of water molecules forming bifurcated hydrogen bonds with starch polymer. At $A_w > 0.6$, there is another transition where the disruption of inter-polymer hydrogen bonds and the formation of water-polymer hydrogen bonds, i.e. bound water, increased to the highest level. The increased number of bound water molecules can be attributed to the unfolding and swelling of starch molecules that resulted in the exposure of more starch polymer sites for water molecules to form hydrogen bonds.

The radius of gyration (R_g) of the starch system was calculated to analyze its overall structural conformation. The calculation was performed for the entire starch system as a whole. The computation involved determining the center of mass (r_{CM}) of the entire starch system, followed by calculating the squared distances of all constituent atoms from r_{CM} , weighted by their respective atomic masses. This provides insight into the spatial distribution of the starch system in water. The swelling and unfolding of starch polymers can be evidenced by the increase in the R_g of the simulated polymers as shown in Fig 5. Coincidentally, both the disruption of inter-polymer hydrogen bonds and the increase of the gyration radius were initiated at much lower hydration levels corresponding to an A_w value of ~ 0.1 and stabilized at an A_w of ~ 0.6 . At the third stage of $A_w > 0.6$, the amount of bound water stabilized, while the amount of unbound water continuously increased. The excess water uptake is associated with the early swelling and the creation of additional free volume in the intermolecular space as observed by the increase in R_g within the starch matrix. The number of water-polymer hydrogen bonds also increased due to the continued increase of a small amount of bifurcated water-polymer hydrogen bonds. While the simulations show swelling and unfolding of starch polymers at high water activities ($A_w > 0.6$), these changes reflect reversible, hydration-driven structural relaxation. This is distinct from gelatinization, which is an irreversible, heat-induced transition involving loss of crystalline order and granular disintegration.

The rigidity of starch at low water activity restricts water mobility not only through strong adsorption (hydrogen bonding) but also by steric confinement and limited free volume. Starch chains are tightly packed, restricting water molecules to diffuse in. As hydration increases, the disruption of hydrogen bonds and swelling of the matrix enhances free volume and chain flexibility, creating new pathways for water transport. Thus, the observed mobility arises from a combination of sorption thermodynamics and polymer dynamics, rather than adsorption energy alone.

The observed stepwise increase in water–starch hydrogen bonds (H-bonds) at $A_w \sim 0.4$ and $A_w \sim 0.6$ suggests three distinct hydration regimes, each inducing specific structural changes in the starch matrix. At $A_w < 0.4$, water molecules primarily interact with surface hydroxyl groups of starch, forming isolated and stable single hydrogen bonds (Fig. 4a). Bifurcated H-bonds are negligible (Fig. 4b) due to limited water availability and minimal structural change of polymer spacing. These H-bonds preserve the chain rigidity. The polymer chain motion is highly restricted, which is consistent with the low radius of gyration (R_g) of polymer (Fig. 5) and low D_{self} of water (Fig. 7).

At $0.4 < A_w < 0.6$, water begins to disrupt intra- and inter-chain starch–starch hydrogen bonds (Fig. 4c), forming new water–starch H-bonds (Fig. 4a). Bifurcated H-bonds begin to increase slightly (Fig. 4b), indicating the formation of water-mediated bridges between adjacent polymer segments. The breakdown of inter-chain hydrogen bonds leads to structural change of initial inter-polymer spacing and local destabilization of the network, evident from the increasing R_g . Water acts as a plasticizer, enhancing polymer mobility and contributing to the increase in D_{self} .

At $A_w > 0.6$, the number of water–starch H-bonds reaches a plateau (Fig. 4a), while bifurcated H-bonds reach their maximum (Fig. 4b), indicating saturation of water–starch interaction sites. Excess water forms a percolating network (Fig. 4d), shifting the hydration environment toward bulk-like dynamics. The starch matrix undergoes further structural change of swelling (sharp rise in R_g in Fig. 5) and inter-polymer chain separation.

Fig. 4d shows that there is close agreement between the amount of bound and unbound water as characterised by the instantaneous number of water-polymer hydrogen bonds at equilibrium and as that characterised by the enduring water-polymer hydrogen bonds.

3.2 Simulated Water Sorption Isotherm

From the total number of bound and unbound water molecules simulated at varying temperatures and A_w values, the water sorption isotherms can be obtained. An example of the resulting isotherm at 40 °C is shown in Fig. 6 together with the experimental data. The experimental data for the isotherm was obtained from the water vapour sorption kinetics of corn starch as shown in Fig.B.1. It exhibits a typical sigmoidal shape with a slower increase at lower A_w values. In this region, water molecules primarily interact with the hydrophilic sites of the starch, such as the hydroxyl groups, through hydrogen bonding. Since the availability of water is limited, most of these interactions occur on the starch surface. Similar findings were reported in a study on maize starch particles by Ociecek et al.(Ociecek et al., 2023), which highlighted the role of hydrophilic sites in initial moisture adsorption.

As the A_w increases, the starch starts to absorb more water. During this phase, water molecules start to penetrate more deeply into the starch polymeric matrix, disrupting the

hydrogen bonds between the starch molecules. The starch polymer chains also start to unfold and swell. Consequently, more free volume is created that allows additional water molecules to penetrate. Studies on modified starch products by Rajabnezhad et al. (Rajabnezhad et al., 2020) further support this observation, indicating that deeper penetration of water molecules enhances swelling and disrupts inter-polymer hydrogen bonds. The experimental and simulated data indicate an accelerated increase in absorbed water content. However, this has not yet led to significant swelling or structural disruption.

When the A_w increases to more than 0.6, accelerated swelling of the polymer accommodates more water molecules to diffuse into the matrix. There is an increased water mobility and more water uptake. Similar results were observed in sorption studies on pea starch edible films by Saberi et al. (Saberi et al., 2015), where high A_w levels led to significant structural changes in starch matrices.

The experimental data at 40 °C match the simulation data closely with some discrepancies that are attributed to the microscopic pores in the real material that cannot be easily simulated using MD. However, overall, the agreement is close, which validates the accuracy of the model in predicting the hydration properties of starch. The simulated results also correlate well with the experimental data of other sorption studies for starch, such as the one by (Peng et al., 2007) at 45 °C, where the water absorption capacity of the starch matrix behaves similarly at higher temperatures.

3.3 Self-Diffusion coefficient of water

MD simulations were conducted to investigate the self-diffusion coefficient, D_{self} , of water within the starch matrix at varying values of A_w and temperature. The simulated self-diffusion coefficient data can be best fitted by the following equation:

$$D_{self} = D_0(1 - e^{-kA_w}) \quad (18)$$

where the self-diffusion in free water, $D_0 = 5.62 \times 10^{-10} \text{ m}^2/\text{s}$ and $k = 5.789$.

As shown in Fig. 7, D_{self} increases considerably but with a reducing positive gradient as A_w increases from 0.1 to 0.6. This observation suggests that, as more water is absorbed by starch, water molecules can move more freely. At low water activities of < 0.2 , the mobility of the water molecules is significantly restricted, due to the tighter interactions of the hydrogen bonds between water and the starch matrix. In this region, the water molecules are closely associated with the hydrophilic sites on the starch molecules, forming hydrogen bonds, which limits their mobility.

For $A_w > 0.6$, D_{self} begins to reach a plateau, suggesting that the water mobility has reached that of free water, $D_0 = 5.53 \times 10^{-10} \text{ m}^2/\text{s}$. The value of D_{self} obtained for free water is consistent with the expected value for the TIP3P water model used in CHARMM36 (Huda et al., 2021). At these high water activities, the starch matrix is sufficiently hydrated and the additional water uptake is primarily due to the increase of inter-polymer spaces. In this region, the molecular mobility of most water molecules is no longer restricted.

3.4 Characteristic sorption parameter and moisture permeability

The slope of the isotherm, dC/dA_w , was obtained by fitting the experimental sorption data to the GAB isotherm equation (Blahovec & Yanniotis, 2008), given by:

$$C = \frac{M_0 A_0 I_0 A_w}{(1 - I_0 A_w)[1 - I_0(1 - A_0)A_w]} \quad (19)$$

where M_0 is the monolayer moisture content, A_0 is a constant related to the interaction energy of water, I_0 is a constant related to the interaction energy of water with the matrix.

The following parameters were obtained by best fitting the GAB equation to the experimental data at 40 °C reported in Part 1: $M_0 = 2.58$ kg of water/ kg of sample, $A_0 = 9.99$, $I_0 = 0.75$. Fig. 8 shows the fitted GAB equation compared to the measured moisture sorption data for corn starch.

The slope of the GAB isotherm equation provides insights into the potential for water absorption at different water activities. As shown in Fig. 9, the system exhibits a characteristic behaviour typical of porous hygroscopic materials (Schmidt, 2020). At low water activities, the potential for absorbing bound water is high due to the availability of hydrogen bonding sites. However, as water activity increases, this potential declines because many of these sites become occupied which reduces the capacity for additional hydrogen bond formation. This trend continues until the slope reaches a minimum value at $A_w = 0.29$, corresponding to the point where most accessible binding sites have been saturated. Beyond this critical point, the potential to absorb water rises again as water molecules begin filling the porous network of the material. This increase in characteristic sorption parameter is further enhanced by the unfolding of starch polymer chains, which expands the porous structure and facilitates additional water uptake.

The characteristic sorption parameter, H , of the starch was derived from the sorption isotherm data, as shown in Fig. 10. At $A_w < 0.4$, H remains relatively small when the starch molecular structure is relatively rigid or compact with minimal swelling. Most absorbed water molecules are bound to starch with hydrogen bonds. The values of H increase significantly in the range $A_w > 0.6$, suggesting that the starch matrix undergoes structural changes. The transition suggests the start of gelatinization or swelling of starch granules, where the molecular structure becomes more open, allowing more water to be absorbed. Swelling leads to the expansion of interchain spaces, which allows for more water uptake.

From the characteristic sorption parameter and measured moisture diffusivity of Part 1, values of the moisture permeability as a function of A_w in the bulk starch have been obtained and are plotted in Fig. 11. The data demonstrate that as the starch becomes more hydrated, it also becomes more permeable to moisture migration.

As the starch absorbs more water, the molecular network of the starch polymer becomes more flexible. This allows greater mobility of moisture through the matrix, which is consistent with the water acting as a plasticizer, reducing the intermolecular forces within the starch and enabling more rapid moisture movement. In addition, the increased water uptake creates additional channels through which water molecules can permeate.

3.5 Correlation between moisture permeability and self-diffusion coefficient

Fig. 12 demonstrates that the moisture permeation through the starch matrix derived from the sorption kinetics data, K , exhibits a clear nonlinear relationship with the self-diffusion coefficient of water, D_{self} , obtained from the MD simulations. The relationship can be described by the following equation:

$$D_{self} = D_0 [1 - \exp(-K/K_0)] \quad (20)$$

where for corn starch, characteristic value of K , $K_0 = 4.42 \times 10^{-14}$ m/s and $D_0 = 5.62 \times 10^{-10}$ m²/s, which indicates that as D_{self} approaches D_0 , there is a rapid increase in K . That is, as the water permeability increases significantly, the self-diffusion coefficient in the starch approaches that of bulk water. The rapid increase in the water permeability corresponds to the region of water activity greater than 0.6 where starch is significantly hydrated, and the self-diffusivity of water in starch approaches the plateau of bulk water. This arises from the swelling of the starch, allowing more channels to be created that enable free water molecules to penetrate through the bulk of the starch.

3.6 Bulk diffusion coefficient

The theoretical model outlined in Eq. (16) was used to predict the bulk diffusion coefficient of water in corn starch. This model was informed by the calculated moisture permeability (from Eq. 20) and the slope of the isotherm (dC/dA_w) obtained from the fitted GAB model (from Eq. 19). Figure 13 shows the comparison of the calculated and experimental bulk diffusion coefficients (D_{bulk}) given in Part 1 with respect to A_w . Overall, the calculated D_{bulk} values follow a similar trend to the experimental data and there is a good agreement. Both experimental data and theoretical model showed the bulk diffusivity initially increase with A_w , reach a maximum, and then decrease at higher water activities. The peak in the calculated D_{bulk} curve occurs around $A_w = 0.3$, which is also suggested by the experimental data.

4. Conclusions

This study investigates the hydration kinetics of starch using Molecular Dynamics (MD) simulations. The simulated water vapor sorption isotherm agrees closely with the experimental data reported in Part 1 as well as the literature. The simulated self-diffusivity of water in starch increases monotonically with the water activity. A simple exponential relationship has been observed between the unsaturated water permeability and the self-diffusivity of water in the starch matrix.

The MD simulations reveal that at low water activities of $A_w < 0.4$, absorbed water molecules are tightly bound to the surface of the starch molecular matrix, mostly forming hydrogen bonds with the surface of the starch molecules. Water mobility is rather restricted in the rigid polymer structures. As the water activity increases in the region between 0.4 and 0.6, absorbed water molecules start to penetrate the inner structures of the starch molecules. Absorbed water becomes more freely mobile, leading to the extension and plasticization of the starch polymer. Both the self-diffusion coefficient and permeability coefficient of water increased rapidly. In addition to the hydration kinetics, structural changes in starch molecules were analysed as a function of water activity. The analysis indicated that starch molecules undergo unfolding and rearrangement with increasing water activity. At water activities

greater than 0.6, absorbed water molecules are mostly freely mobile in the starch matrix with the self-diffusivity of the water approaching the plateau value for pure water. However, the unfolding and swelling of the starch polymer chains continue in this region with a continued increase in the inter-polymer space, the water channels and the water holding capacity. The consequence is a rapidly increasing moisture permeability in this region. It was also observed across the three hydration stages, low A_w (<0.4), intermediate A_w (0.4 to 0.6), and high A_w (>0.6), that they have distinctive water-polymer interactions. While bifurcated water-polymer hydrogen-bonds increased with increasing A_w , the number is much smaller than that of single water-polymer hydrogen bonds. The number of water-polymer hydrogen-bonded water molecules remained almost constant in each stage and increased from each stage. For $A_w < 0.4$, water molecules are mostly bound to the exposed starch polymer surface. In the intermediate region of $0.4 < A_w < 0.6$, more water molecules penetrated the inter-polymer space and form hydrogen bonds. In the high water activity region, the swelling and unfolding of starch polymers expose additional sites for water molecules to form water-polymer hydrogen bonds

An important finding of the MD simulation is that both the self-diffusivity and permeability of water in the starch matrix increase with water activity monotonically. The observed decreases in the bulk diffusion coefficient at an $A_w > 0.4$ is due to the rapid increase in the characteristic sorption parameter caused by polymer extension and swelling. It is not an indication of the reduction in water mobility. The predictions of the MD simulations agreed closely with the experimental data reported in Part 1.

This study provides molecular-level insights into water-starch interactions across different water activities. It helps to improve starch-based materials used in food and pharmaceuticals by understanding the hydration dynamics that can affect the stability, texture, and performance of these products.

Author contributions: CRediT

Edwin Joseph: Methodology, Software, Validation, Formal analysis, Investigation, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization. **Guoping Lian:** Conceptualization, Methodology, Formal analysis, Resources, Writing - Review & Editing, Supervision, Funding acquisition. **Michael J. Adams:** Conceptualization, Resources, Writing - Review & Editing, Supervision, Project administration, Funding acquisition. **Zhibing Zhang:** Validation. **Andrew T. Tyowua:** Validation. **Chuan-Yu Wu:** Conceptualization, Resources, Writing - Review & Editing, Supervision, Project administration, Funding acquisition.

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Conflict of interests

The authors declare that they have no competing interests.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

All the data points corresponding to the figures presented in this study are provided as supplementary information in an excel file.

Appendix B. Supplementary data

All experimental data relevant to this work from Part 1 of the study are provided as supplementary information for reference and completeness.

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